4	**(a)**	**(i)** $T = 0.60\,\text{s}$ <u>and</u> $\omega = 2\pi / T$	C1	
		$\omega = 10\,(10.47)\ \text{rad s}^{-1}$	A1	[2]
		(ii) energy $= \tfrac{1}{2}m\omega^2 x_0^2$ or $\tfrac{1}{2}mv^2$ <u>and</u> $v = \omega x_0$	C1	
		$= \tfrac{1}{2} \times 120 \times 10^{-3} \times (10.5)^2 \times (2.0 \times 10^{-2})^2$		
		$= 2.6 \times 10^{-3}\ \text{J}$	A1	[2]
	(b)	sketch: smooth curve in correct directions	B1	
		peak at f	M1	
		amplitude never zero and line extends from $0.7f$ to $1.3f$	A1	[3]
	(c)	sketch: peaked line always below a peaked line A	M1	
		peak not as sharp <u>and</u> at (or slightly less than) frequency of peak in line A	A1	[2]